TASK24: Vibe Cloud App

AIM: To develop an application that uses iterative prompt engineering to generate and evaluate REST APIs, cloud deployment steps, and security configurations.

ALGORITHM:

1. Start the program
2. Create project files (HTML, CSS, JS, server.js, db.js)
3. Design UI to accept user prompts
4. Capture prompt input using JavaScript
5. Send prompt to backend using fetch API
6. Receive prompt in server and forward to processing module
7. Store prompt in history for version tracking
8. Analyze prompt type (REST API / Cloud / Security)
9. Generate corresponding output based on prompt
10. Evaluate output quality and assign score
11. Return output and score to frontend and display results
12. Stop the program

PROGRAM:
INDEX.HTML:

```
<!DOCTYPE html>

<html>

<head>

   <title>Vibe Coding App</title>

   <link rel="stylesheet" href="style.css">

</head>

<body>

<div class="container">

   <h2>Vibe Coding Cloud Generator</h2>
```

```
    <textarea id="prompt" placeholder="Enter prompt..."></textarea>
    <button onclick="generate()">Generate</button>
    <h3>Output</h3>
    <pre id="output"></pre>
    <h3>Score</h3>
    <p id="score"></p>
    <h3>Prompt Versions</h3>
    <ul id="history"></ul>
</div>
<script src="script.js"></script>
</body>
</html>
```

STYLE.CSS:

```
body {
    font-family: Arial;
    background: #b98bdb;
    color: white;
    display:flex;
    justify-content:center;
    align-items:center;
    height:100vh;
}
.container {
    width: 400px;
    text-align: center;
}
```

```
textarea {
    width: 100%;
    height: 100px;
}
button {
    padding: 10px;
    margin: 10px;
    background: #00c6ff;
    border: none;
}
pre {
    background: #333;
    padding: 10px;
}
li {
    background: #444;
    margin: 5px;
    padding: 5px;
}
```

SCRIPT.JS:

```
function generate() {
    const prompt = document.getElementById("prompt").value;
    fetch("/generate", {
        method: "POST",
        headers: {"Content-Type":"application/json"},
```

```
        body: JSON.stringify({ prompt })
    })
    .then(res => res.json())
    .then(data => {
        document.getElementById("output").innerText = data.output;
        document.getElementById("score").innerText = data.score + "/10";
        loadHistory();
    });
}
function loadHistory() {
    fetch("/history")
    .then(res => res.json())
    .then(data => {
        let list = document.getElementById("history");
        list.innerHTML = "";
        data.forEach((p, i) => {
            let li = document.createElement("li");
            li.innerText = `v${i+1}: ${p.prompt}`;
            list.appendChild(li);
        });
    });
}
loadHistory();
```

SERVER.JS:

```
const express = require("express");
const db = require("./db");
```

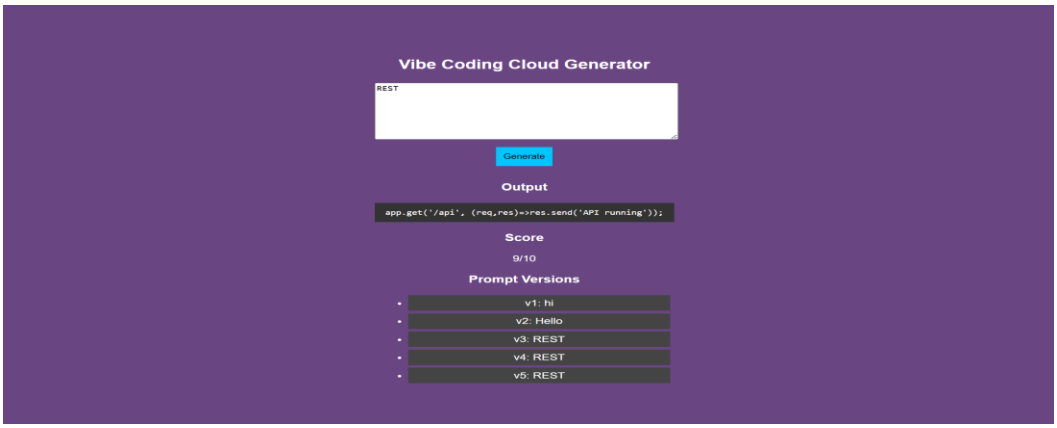
```
const app = express();
app.use(express.json());
app.use(express.static(__dirname));
app.post("/generate", (req, res) => {
  const result = db.process(req.body.prompt);
  res.json(result);
});
app.get("/history", (req, res) => {
  res.json(db.getHistory());
});
app.listen(3000, () => {
  console.log(" 🚀 http://localhost:3000");
});
```

DB.JS:

```
let history = [];
function process(prompt) {
  let output = "";
  let score = 5;
  history.push({ prompt });
  if (prompt.includes("REST")) {
    output = `app.get('/api', (req,res)=>res.send('API running'));`;
    score = 7;
  }
  else if (prompt.includes("cloud")) {
    output = `Deploy using AWS EC2:
1. Launch instance
```

```
2. Install Node.js

3. Run app`;

        score = 8;

    }

    else if (prompt.includes("security")) {

        output = `Use HTTPS, JWT authentication, and firewall rules`;

        score = 9;

    }

    score += Math.min(history.length, 2);

    return { output, score };

}

function getHistory() {

    return history;

}

module.exports = { process, getHistory };
```

OUTPUT:



RESULT: A Vibe Coding application was successfully developed to generate and refine cloud-based features using iterative prompt engineering.